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GCU Interface Specification

MODEL : GCU
 REV. : 2.8
 DATE : 2023. 11.23



PULOON Technology Inc.

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Revision History

Ver.	DATE	Item		Name
		Title	Details	
1.2	2018.04.19	First Released	Separate interface specification	
1.3	2018.04.25	Add SetRegion command	Add SetRegion Command	
1.4	2019.01.25	Add Child Detection Option	Add Child Detection Option	
1.5	2019.05.08	Add SwingDoor Option	Add SwingDoor specification	
1.6	2019.07.17	Fix Chapter 3.2, 4.3		
1.7	2019.09.26	Change sensor number		
1.8	2019.11.14	Modify PassMode	Modify PassMode	
1.9	2020.03.02	Modify Chapter. 4.3.2, 4.3.7	Modify Option code and Add RTC, UPS Command.	
2.0	2020.07.28		Modify Region Area.	
2.1	2020.08.31	Modify Chapter. 4.3.12	Modify parameter setting, UPS	
2.2	2021.04.26	Modify Chapter. 4.3.5	Add Test Passage Mode	
2.3	2021.04.30	Modify Chapter 4.3.11	Add UPS charge level	
2.4	2022.04.20	Modify Chapter 4.3.7	Add CRC16 calculation method	
2.5	2022.07.28	Add Appendix A.	Add TCU option	
2.6	2022.11.23	Modify Chapter 4.3.2, 4.3.7	Midify Sub2 data example	
2.7	2023.11.16	Modify Chapter 4.3.2	Modify Chapter 4.3.2	
		Add Chapter 4.3.15	Add Chapter 4.3.15	
		Add Chapter 4.3.16	Add Chapter 4.3.16	
		Modify Chapter 4.3.8, 4.3.10	Modify Chapter 4.3.8, 4.3.10	
2.8	2023.11.23	Modify Chapter 4.3.7, Modify Chapter 4.3.15 Modify Chapter 4.3.16	Modify content in the chapter	

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1 Overview

This document describes the Interface Specification of the GCU Board.

2 Communication interface

2.1 General Description

Communication specification is as follows.

Item	Specification
Communication Method	RS232, Half duplex
Communication Speed (bps)	57600 (default)
Data Bit	8 bits
Parity Bit	No Parity
Stop Bit	1 bit

2.2 Pin Assignment

Pin assign for communication connection is as follows.

Pin Description (Connector Maker Pin No)				
Pin No	1	2		4
Description	TXD	RXD		GND

2.3 Code Description

The control code used for communication is as follows.

Code	Value	Description
LF	0Ah	Start of Text
CR	0Dh	End of Text

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3 Message Frame

The configuration of the packet (Message Frame) is as follows. It consists of an ASCII code between the start code LF and the end code CR.

3.1 Message Format

Code	(1) LF	(2) SEQ. No	(3) Retry No.	(4) DATA	(5) CRC16	(6) CR
Bytes	1	2	1	Variable	4	1
Type	HEX	HEX	HEX	ASCII	HEX	HEX

3.2 Error Check Code(CRC-16)

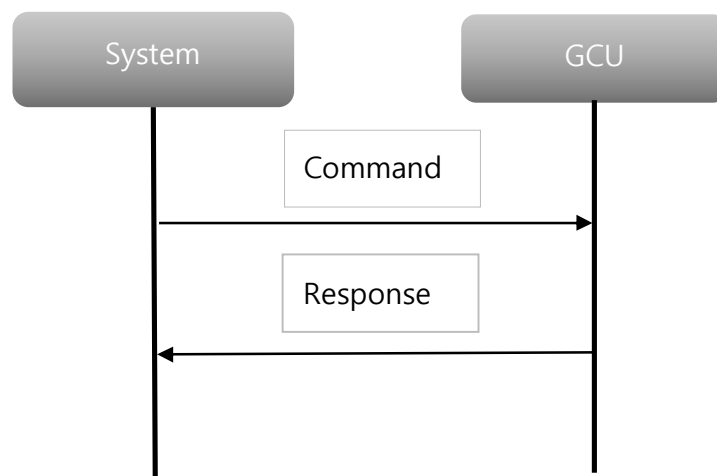
The message contains Error Check Code based on CRC-16 method. The Error Check Code consists of 32 bits (4 bytes). The receiver recomputes the CRC in the received message and compares the result with the actually received Error Check Code. The CRC calculation range is from SEQ.No to DATA (ErrorCode). If the CRC values do not match, send a error. For the crc16 calculation method, refer to Appendix A.

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4 Communication Flow Chart

4.1 Normal Flow

The communication between the system and the GCU is to send a response whenever a command is received. During Status Polling, the Interval should be over 100ms.



4.2 Data Definitions in Command

The Data field consists of Command and Sub Data.

DATA	
CMD	Sub Data
1 Byte	(1-N) Bytes

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4.3 Command List

Command List is as follows:

Category	Code	Description	Remark
Passage Allowed	'A'	Pass authentication signal, Allow entry or exit.	
Count Clear	'C'	Clear Entry and Exit Count and close the door.	
Set Pass Mode	'D'	Set the Pass Mode of the door.	
Emergency	'E'	Change gate to emergency mode, Allow all pass.	
Set Region	'G'	Set the Security, Safety area.	
Sensor Status	'H'	Send the Sensor Status.	
Initialization	'I'	Initialize the GCU.	
Parameter	'P'	Set the GCU parameters.	
Reset	'R'	H/W Reset the GCU.	
Status	'S'	Send the Status.	
Test	'T'	Door, direction indicator, buzzer, etc. are tested.	
Version	'V'	Send the F/W Version.	
DateTime Setting	'X'	Set DateTime of RTC. (for India)	
UPS	'Y'	Control the UPS power. (for India)	

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4.3.1 Version

Send the version of the firmware. The version is 5 bytes total, consisting of Major Version 2Bytes, 1Byte of dot (.), And Minor Version 2Bytes.

(1) Command

STX	Seq No.	Retry No.	DATA	Error Check Code	ETX
LF	Sequence	Retry	CMD	CRC16	CR
0x0A			0x56		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	4 Bytes	1 Byte

(2) Response

STX	Seq No.	Retry No.	DATA			Error Check Code	ETX
LF	Sequence	Retry	CMD	Error Code	Sub Data	CRC16	CR
0x0A			0x56	00	XX.YY		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes	5 Bytes	4 Bytes	1 Byte

* Sub Data Details

XX: Major Version Number

YY: Minor Version Number

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4.3.2 Passage Allowed

Only one pass is allowed. Select the entry or exit direction and send. Pass the number of allowed commands can be cumulative. If you send Entry Passage Allowed twice, you can enter 2 people at the gate before timeout. If Entry, Exit commands are sent, Exit is permitted after Entry is completed.

(1) Command

STX	Seq No.	Retry No.	DATA		Error Check Code	ETX
LF	Sequence	Retry	CMD	Sub Datas	CRC16	CR
0x0A			0x41	*		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	1 or 4 Bytes	4 Bytes	1 Byte

* SubDatas : Only India used 4Bytes, 1Byte for all other sites.

* SubData 1 <Passage Direction> - 1 Byte

0x30: Entry Pass

0x31: Exit Pass

* SubData 2 <Top-lamp color> - 1 Byte(Only used India)

- Valid Ticket

0x30 + Color combination value

combination value: 0x01 - Green, 0x02 - Blue, 0x04 - Red

Ex) Yellow: 0x35(0x30 + 0x01 + 0x04), off: 0x30

- Invalid Ticket(Only the top lamp of Passage Direction is displayed in the designated color for 0.5 seconds - Only used BldcSectorGCU)

0x40 + Color combination value

combination value: 0x01 - Green, 0x02 - Blue, 0x04 - Red(default)

In case of invalid ticket, set SubData3 to 0x00.

Ex) Red: 0x44(0x30 + 0x04)

* SubData 3 <Passage count> - 2 Byte(Only used India)

Range: 01 ~ 99

Ex) 10 passengers = 10(0x31, 0x30), 15 passengers = 15(0x31, 0x35)

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(2) Response

STX	Seq No.	Retry No.	DATA		Error Check Code	ETX
LF	Sequence	Retry	CMD	Error Code	CRC16	CR
0x0A			0x41	00		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes	4 Bytes	1 Byte

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4.3.3 Emergency

This is a command to set the gate to the emergency mode or releases the emergency mode. It operates the same as the emergency mode signal input by the emergency signal. When the gate enters the emergency mode, the door is opened, and the direction indicator turns on the blue arrow, which signifies passage. And it does not detect the sensor. All passengers can pass through the gate without interruption. When emergency mode is OFF, it returns to the previously set pass mode.

(1) Command

STX	Seq No.	Retry No.	DATA		Error Check Code	ETX
LF	Sequence	Retry	CMD	Sub Data	CRC16	CR
0x0A			0x45	*		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	1 Byte	4 Bytes	1 Byte

* Sub Data Details

0x31: Emergency ON

0x30: Emergency OFF

(2) Response

STX	Seq No.	Retry No.	DATA		Error Check Code	ETX
LF	Sequence	Retry	CMD	Error Code	CRC16	CR
0x0A			0x45	00		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes	4 Bytes	1 Byte

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4.3.4 Initialization

Initialize the GCU. During initialization, sensor check, side door switch check, buzzer initialization, direction indicator initialization and door initialization are performed.

(1) Command

STX	Seq No.	Retry No.	DATA	Error Check Code	ETX
LF	Sequence	Retry	CMD	CRC16	CR
0x0A			0x49		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	4 Bytes	1 Byte

(2) Response

STX	Seq No.	Retry No.	DATA		Error Check Code	ETX
LF	Sequence	Retry	CMD	Error Code	CRC16	CR
0x0A			0x49	00		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes	4 Bytes	1 Byte

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4.3.5 Parameter

(1) Command

STX	Seq No.	Retry No.	DATA			Error Check Code	ETX
LF	Sequence	Retry	CMD	Sub Data	Parameters	CRC16	CR
0x0A			0x50	*	*		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	1 Byte	0 or 12 Bytes	4 Bytes	1 Byte

* Sub Data Details

0x30: Parameter Set(Parameters length 12 Bytes)

0x31: Parameter Get(Parameters length zero)

* Paramter 1 <No Entry Time Out> - 3 Bytes

Time allowed for passage. After the set time has elapsed, the buzzer sounds and the door closes. The setting unit is 1 sec.

ex 1) 10 sec = 010(0x30, 0x31, 0x30)

ex 2) 15 sec = 005(0x30, 0x31, 0x35)

* Paramter 2 <Gate Mode> - 1 Byte

0x30: Normal Open(Not used in SwingDoor)

0x31: Normal Close

* Paramter 3 <Hurry up Entry Permission Mode> - 1 Byte

Set the area where the buzzer does not sound even when entering the gate without Passage Allowed (TAG).

This option is used when TAG after entering the gate first.

a) In the case of SectorDoor

0x30: None

0x31: Level 1 (S12 / S18)

0x32: Level 2 (S2, S3, S5 / S18, S17, S15)

0x33: Level 3 (S2, S3, S5, S1, S6, / S18, S17, S15, S14, S19)

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b) In the case of SwingDoor

0x30: None

0x31: Level 1 (S1 / S11)

0x32: Level 2 (S1, S2, S3 / S11, S12, S13)

0x33: Level 3 (S1, S2, S3, S4, S5 / S11, S12, S13, S14, S15)

* Paramter 4 <RF Tag Timeout Mode> - 1 Byte

0x30: Each Tag Time

No Entry Timeout time increases for each TAP.

When the No Entry timeout is 15 sec, if TAG is executed three times, the timeout will be after 45 sec ($15 + 15 + 15 = 45$).

After TAG three times, if 1 person passes through the gate, it is reduced by 15 sec for 1 person, leaving 30 sec.

0x31: Stack Last Tag Time

The No Entry Timeout time is updated based on the last TAG time.

When the No Entry timeout time is 15 sec, even if the TAG is performed three times, the timeout is 15 sec after the last TAG.

* Paramter 5 <Tailing Performance> - 1 Byte

0x30: Hi

0x31: Low

* Parameter 6 <Buzzer Timeout> - 2 Bytes

Set the timeout value of the buzzer. In a situation where an alarm has occurred, the buzzer sounds for the set time.

The default 5 sec and the setting value are added. The range is 0x00~0xFE and the setting value unit is 100 msec.

To make the buzzer sound for 20 sec, it should be set to 96h. $20 \text{ sec} = 5 \text{ sec} + 15 \text{ sec}$ ($96\text{h} = 150 * 100 \text{ msec}$)

default value: 64h = 5sec + 10sec($100*100\text{msec}$)

Buzzer Timeout(15 sec): default 5 sec + 10($100*100\text{msec}$) sec

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* Parameter 7 <Safety Region Timeout> - 2 Bytes

For passenger safety, even after No Entry Timeout, the door remains open when the passenger is in the safety zone. After the time set as the Safety Region Timeout, the door will be closed even if there are passengers in the safety area.

The setting unit is 1 sec and range is 0x00 ~ 0xFF(0xFF: Disable Timeout). When set to FFh, the door will not be closed indefinitely if the passenger is in the safe area.

If set to 1Eh, the door will not be closed for 30 sec if the passenger is in a safe zone after the No Entry Timeout. After 30 sec, the door closes even if the passenger is in the safe area.

default 값: 1E(h) = 30 sec

* Parameter 8 <Child Detect Function On/Off> - 1 Byte

0x30: Child Detection Sensor disable(default value in SwingDoor)

0x31: Child Detection Sensor enable Level 1(default value in SectorDoor)

0x32: Child Detection Sensor enable Level 2(Optional)

※ Enable value(0x31, 0x32) is available only when a separate child sensor is installed.(for China)

In India, It should always be set to Disable(0x30).

(2) Response

STX	Seq No.	Retry No.	DATA				Error Check Code	ETX
LF	Sequence	Retry	CMD	Error Code	Sub Data	Parameters	CRC16	CR
0x0A			0x50	00	*	*		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes	1 Byte	12 Bytes	4 Bytes	1 Byte

* Sub Data

Same as Sub Data format of Command

* Parameters

Same as Parameters 12Bytes format of Command

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4.3.6 Reset

It is a H/W reset command of the GCU board.

(1) Command

STX	Seq No.	Retry No.	DATA	Error Check Code	ETX
LF	Sequence	Retry	CMD	CRC16	CR
0x0A			0x52		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	4 Bytes	1 Byte

(2) Response

STX	Seq No.	Retry No.	DATA		Error Check Code	ETX
LF	Sequence	Retry	CMD	Error Code	CRC16	CR
0x0A			0x52	00		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes	4 Bytes	1 Byte

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4.3.7 Status

It sends Status such as the sensor, a side door switch, DIP switch.

(1) Command

STX	Seq No.	Retry No.	DATA	Error Check Code	ETX
LF	Sequence	Retry	CMD	CRC16	CR
0x0A			0x53		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	4 Bytes	1 Byte

(2) Response

STX	Seq No.	Retry No.	DATA			Error Check Code	ETX
LF	Sequence	Retry	CMD	Error Code	Sub Data	CRC16	CR
0x0A			0x53	00	*		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes	23 or 27 Bytes	4 Bytes	1 Byte

* Sub Data Length: 27 Bytes when using UPS, 23 Bytes for all other sites.

* Sub Data Detail List

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No	Size(Bytes)	Name	Descriptions
1	1	Door Running State	0x3D: Maintenance Mode 0x30: Entry/Exit is Contolled 0x31: Entry is Free Pass, Exit is Locked 0x32: Exit is Free Pass, Entry is Locked 0x33: Entry is Controlled, Exit is Locked 0x34: Exit is Controlled, Entry is Locked 0x35: Entry is Free Pass, Exit is Controlled 0x36: Exit is Free Pass, Entry is Controlled 0x37: Entry is Free Pass, Exit is Controlled(Not used in SwingDoor) 0x38: Exit is Free Pass, Entry is Controlled(Not used in SwingDoor) 0x39: Entry is Free Pass, Exit is Locked 0x3A: Exit is Free Pass, Entry is Locked 0x3B: Entry is Free Pass, Exit is Free Pass 0x3C: Entry is Locked, Exit is Locked 0x3D: Maintenance Mode 0x3E: Test Passage Mode(only used India) 0x3F: Out of service Mode(only used India)
2	2	Enter Passage Count	00 – 99
3	2	Exit Passage Count	00 – 99

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4	1	Reason of count deduced	0x30: Idle 0x31: Entry passage passing complete 0x32: Exit passage passing complete 0x33: Entry passage timeout(No one at the gate) 0x34: Exit passage timeout(No one at the gate) 0x35: Entry passage passingtimeout(Before door) 0x36: Exit passage passingtimeout(Before door) 0x37: Entry passage tailgate used 0x38: Exit passage tailgate used - Door is closed because of Tailing, but person was already passed the door. This case: the Count is decreased and Return to the Normal wait status 0x39: Entry passage wrongway used 0x40: Exit passage wrongway used - Door is closed because of Wrongway, but person was already passed the door. This case: the Count is decreased and Return to the Normal wait status
5	1	Entry Error Code	0x30: Normal 0x33: Piggy Tailing 0x35: Intrusion 0x39: Wrong way fraud
6	1	Exit Error Code	0x30: Normal 0x33: Piggy Tailing 0x35: Intrusion 0x39: Wrong way fraud
7	1	Door Error Code	0x40: Default (Each Bit Set by each error) 0x01: Door1 Open Fail 0x02: Door1 Close Fail 0x04: Door2 Open Fail 0x08: Door2 Close Fail
8	1	Gate Inner State	0x80: Default (Each Bit set by detected state) 0x01: Detected on the block 1 0x02: Detected on the block 2 0x04: Detected on the block 3 0x08: Detected on the safety area 0x10: Detected on the oposit side block 3 0x20: Detected on the oposit side block 2 0x40: Detected on the oposit side block 1

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9	3	Switch Status	0x40 – 0x4F: Last 4 bit means each input On/Off First Byte 0x01: Door Switch 11 On/Off 0x02: Door Switch 12 On/Off 0x04: Door Switch 13 On/Off 0x08: Door Switch 21 On/Off Second Byte 0x01: Door Switch 22 On/Off 0x02: Door Switch 23 On/Off 0x04: Door Switch 31 On/Off 0x08: Door Switch 32 On/Off Third Byte 0x01: Door Switch 33 On/Off 0x02: Door Switch 41 On/Off 0x04: Door Switch 42 On/Off 0x08: Door Switch 43 On/Off
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10	8	Input Status	0x30 – 0x3F: Last 4 bit means each input On/Off <For China> Third Byte 0x01: Entry Signal Input On/Off 0x02: Exit Signal Input On/Off <For India> Third Byte 0x02: Cabinet Switch 2 On/Off 0x04: Cabinet Switch 3 On/Off 0x08: Cabinet Switch 4 On/Off Fourth Byte 0x01: Cabinet Switch 5 On/Off 0x02: Cabinet Switch 6 On/Off 0x04: Cabinet Switch 7 On/Off Fifth Byte 0x01: Spare Switch 1 On/Off 0x02: Spare Switch 2 On/Off 0x04: Spare Switch 3 On/Off 0x08: Spare Switch 4 On/Off Sixth Byte 0x01: Spare Switch 5 On/Off 0x02: Spare Switch 6 On/Off 0x04: Spare Switch 7 On/Off 0x08: Spare Switch 8 On/Off Seventh Byte - When the TCU is not in use 0x01: Spare Input 1 On/Off 0x02: Spare Input 2 On/Off 0x04: Spare Input 3 On/Off 0x08: Spare Input 4 On/Off - When the TCU is in use 0x01: Spare Input 1 On/Off 0x02: Token Box A Switch On/Off 0x04: Token Box B Switch On/Off 0x08: Spare Input 4 On/Off Eighth Byte 0x01: Spare Input 5 On/Off 0x02: Spare Input 6 On/Off 0x04: Spare Input 7 On/Off 0x08: Spare Input 8 On/Off
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11	1	Emergency Status	0x30: None 0x31: Emergency state by ECU 0x32: Emergency state by 24V 0x33: Emergency state by ECU and 24V
12	1	Sensor Error	0x30 : No Error 0x31 : sensor error 0x32 : child sensor error
13	4	UPS Status (Used only when connected to UPS in India.)	First Byte 0x08: OnLine Mode Second Byte 0x01: OnBattery Mode Third~Fourth Bytes Charging level: 00 ~ 99 percent, FF: Unknown 00(0x30 0x30) 99(0x39 0x39)
14	2	Token Path A	00~99(0x30 0x30 ~ 0x39 0x39) Initialize after response
15	2	Token Path B	00~99(0x30 0x30 ~ 0x39 0x39) Initialize after response
16	2	Token Path ReturnCup	00~01(0x30 0x30 ~ 0x30 0x31) Initialization after response in return cup clear state

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4.3.8 Test

The GCU tests Various controllable modules, the doors for opening, closing, free locking, direction indicator control, and buzzer sounding.

(1) Command

STX	Seq No.	Retry No.	DATA		Error Check Code	ETX
LF	Sequence	Retry	CMD	Sub Data	CRC16	CR
0x0A			0x54	*		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes	4 Bytes	1 Byte

* Sub Data 1 Details

0x30: Door Test

0x31: End Display Test

0x32: Indicator Test

0x33: Buzzer Test

0x35: Lamp Test

* Sub Data 2 Details if Sub Data 1 is Door Test

0x31: Door Open Test

0x32: Door Close Test

0x33: Door Free Test

0x34: Door Lock Test

* Sub Data 2 Details if Sub Data 1 is End Display Test

0x31: End Display Green ON

0x32: End Display Green OFF

0x33: End Display Yellow ON

0x34: End Display Yellow OFF

0x35: End Display Red ON

0x36: End Display Red OFF

* Sub Data 2 Details if Sub Data 1 is Indicator(ConcessionLamp) Test

0x31: Indicator Green ON

0x32: Indicator Green OFF

0x33: Indicator Blue ON

0x34: Indicator Blue OFF

0x35: Indicator Red ON

0x36: Indicator Red OFF

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* Sub Data 2 Details if Sub Data 1 is Buzzer Test

0x31: Buzzer 1 ON

0x32: Buzzer 1 OFF

0x33: Buzzer 2 ON

0x34: Buzzer 2 OFF

0x35: Buzzer 3 ON

0x36: Buzzer 3 OFF

* Sub Data 2 Details if Sub Data 1 is Lamp Test

0x37: ReturnCup LED ON

0x38: ReturnCup LED OFF

(2) Response

STX	Seq No.	Retry No.	DATA		Error Check Code	ETX
LF	Sequence	Retry	CMD	Error Code	CRC16	CR
0x0A			0x54	00		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes	4 Bytes	1 Byte

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4.3.9 Passage Clear

Clear the Pass count.

(1) Command

STX	Seq No.	Retry No.	DATA	Error Check Code	ETX
LF	Sequence	Retry	CMD	CRC16	CR
0x0A			0x43		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	4 Bytes	1 Byte

(2) Response

STX	Seq No.	Retry No.	DATA		Error Check Code	ETX
LF	Sequence	Retry	CMD	Error Code	CRC16	CR
0x0A			0x43	00		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes	4 Bytes	1 Byte

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4.3.10 Sensor Status

Sensor On/Off Status and Error Check

(1) Command

STX	Seq No.	Retry No.	DATA	Error Check Code	ETX
LF	Sequence	Retry	CMD	CRC16	CR
0x0A			0x48		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	4 Bytes	1 Byte

(2) Response

STX	Seq No.	Retry No.	DATA			Error Check Code	ETX
LF	Sequence	Retry	CMD	Error Code	Sub Data	CRC16	CR
0x0A			0x48	00	*		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes	12 Bytes	4 Bytes	1 Byte

* Sub Data Details

0x30 – 0x3F: Last 4 bit means each sensor On/Off

No	Size(Byte)	Name	Descriptions
1	1	Sensor Status First Byte	0x01: Sensor 01 On/Off 0x02: Sensor 02 On/Off 0x04: Sensor 03 On/Off 0x08: Sensor 04 On/Off
2	1	Sensor Status Second Byte	0x01: Sensor 05 On/Off 0x02: Sensor 06 On/Off 0x04: Sensor 07 On/Off 0x08: Sensor 08 On/Off

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3	1	Sensor Status Third Byte	- In case of using SectorDoor 0x01: Sensor 09 On/Off 0x02: always Off 0x04: Sensor 11 On/Off 0x08: Sensor 12 On/Off - When using TCU(In case of using SectorDoor) 0x01: Sensor 09 On/Off 0x02: TCU A Box Path Sensor On/Off 0x04: Sensor 11 On/Off 0x08: Sensor 12 On/Off - In case of using SwingDoor 0x01: Sensor 09 On/Off 0x02: Sensor 23 On/Off 0x04: Sensor 11 On/Off 0x08: Sensor 12 On/Off
4	1	Sensor Status Fourth Byte	0x01: Sensor 13 On/Off 0x02: Sensor 14 On/Off 0x04: Sensor 15 On/Off 0x08: Sensor 16 On/Off
5	1	Sensor Status Fifth Byte	- In case of using SectorDoor 0x01: Sensor 17 On/Off 0x02: Sensor 18 On/Off 0x04: Sensor 19 On/Off 0x08: always Off - When using TCU(In case of using SectorDoor) 0x01: Sensor 17 On/Off 0x02: Sensor 18 On/Off 0x04: Sensor 19 On/Off 0x08: TCU B Box Path Sensor On/Off - In the case of SwingDoor 0x01: Sensor 17 On/Off 0x02: Sensor 18 On/Off 0x04: Sensor 19 On/Off 0x08: Sensor 24 On/Off

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6	1	Sensor Status Sixth Byte	<For China> If Child Detect Off then always 0x3f, Else 0x01: Sensor 10(Child Detect1) On/Off 0x02: Sensor 20(Child Detect2) On/Off 0x04: Sensor 21(Child Detect3) On/Off 0x08: Sensor 22(Child Detect4) On/Off <For India> If When using TCU then 0x01: always On 0x02: always On 0x04: Sensor 21(Return cup sensor) On/Off 0x08: Sensor 22(Return cup sensor) On/Off Else always 0x3f
7	1	Sensor Error Status First Byte	0x01: Sensor 01 no error/error 0x02: Sensor 02 no error/error 0x04: Sensor 03 no error/error 0x08: Sensor 04 no error/error
8	1	Sensor Error Status Second Byte	0x01: Sensor 05 no error/error 0x02: Sensor 06 no error/error 0x04: Sensor 07 no error/error 0x08: Sensor 08 no error/error
9	1	Sensor Error Status Third Byte	- In case of using SectorDoor 0x01: Sensor 09 no error/error 0x02: always Off 0x04: Sensor 11 no error/error 0x08: Sensor 12 no error/error - When using TCU(In case of using SectorDoor) 0x01: Sensor 09 no error/error 0x02: TCU A Box Path Sensor no error/error 0x04: Sensor 11 no error/error 0x08: Sensor 12 no error/error - In case of using SwingDoor 0x01: Sensor 09 no error/error 0x02: Sensor 23 no error/error 0x04: Sensor 11 no error/error 0x08: Sensor 12 no error/error

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10	1	Sensor Error Status Fourth Byte	0x01: Sensor 13 no error/error 0x02: Sensor 14 no error/error 0x04: Sensor 15 no error/error 0x08: Sensor 16 no error/error
11	1	Sensor Error Status Fifth Byte	- In case of using SectorDoor 0x01: Sensor 17 no error/error 0x02: Sensor 18 no error/error 0x04: Sensor 19 no error/error 0x08: always Off - When using TCU(In case of using SectorDoor) 0x01: Sensor 17 no error/error 0x02: Sensor 18 no error/error 0x04: Sensor 19 no error/error 0x08: TCU B Box Path Sensor no error/error - In case of using SwingDoor 0x01: Sensor 17 no error/error 0x02: Sensor 18 no error/error 0x04: Sensor 19 no error/error 0x08: Sensor 24 no error/error
12	1	Sensor Error Status Sixth Byte	<For China> If Child Detect Off then always 0x30, else 0x01: Sensor 10(Child Detect1) no error/error 0x02: Sensor 20(Child Detect2) no error/error 0x04: Sensor 21(Child Detect3) no error/error 0x08: Sensor 22(Child Detect4) no error/error <For India> If When using TCU then 0x01: always no error 0x02: always no error 0x04: Sensor 21(Return cup sensor) no error/error 0x08: Sensor 22(Return cup sensor) no error/error Else always 0x30

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4.3.11 Set Pass Mode

Set the Gate Pass Mode

(1) Command

STX	Seq No.	Retry No.	DATA		Error Check Code	ETX
LF	Sequence	Retry	CMD	Sub Data	CRC16	CR
0x0A			0x44	*		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	1 Byte	4 Bytes	1 Byte

* Sub Data Details

0x30: Entry/Exit is Controlled(SwingDoor used only Normal Close mode)

0x31: Entry is Free Pass, Exit is Locked(Used only Normal Close mode)

0x32: Exit is Free Pass, Entry is Locked(Used only Normal Close mode)

0x33: Entry is Controlled, Exit is Locked

0x34: Exit is Controlled, Entry is Locked

0x35: Entry is Free Pass, Exit is Controlled(SwingDoor used only Normal Close mode)

0x36: Exit is Free Pass, Entry is Controlled(SwingDoor used only Normal Close mode)

0x37: Entry is Free Pass, Exit is Controlled(DoorMode: Normal Open, Not used in SwingDoor)

0x38: Exit is Free Pass, Entry is Controlled(DoorMode: Normal Open, Not used in SwingDoor)

0x39: Entry is Free Pass, Exit is Locked(Used only Normal Open mode)

0x3A: Exit is Free Pass, Entry is Locked(Used only Normal Open mode)

0x3B: Entry is Free Pass, Exit is Free Pass(SectorDoor used only Normal Open mode , SwingDoor used only Normal Close mode)

0x3C: Entry is Locked, Exit is Locked

0x3D: Maintenance Mode

0x3E: Test Passage mode(Only used India)

0x3F: Out of service mode(Only used India)

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(2) Response

STX	Seq No.	Retry No.	DATA		Error Check Code	ETX
LF	Sequence	Retry	CMD	Error Code	CRC16	CR
0x0A			0x44	00		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes	4 Bytes	1 Byte

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4.3.12 Set Region

Set the safety area sensor range. Do not close the door if the sensor is detected in the safety area set by Set Region.

(1) Command

STX	Seq No.	Retry No.	DATA		Error Check Code	ETX
LF	Sequence	Retry	CMD	Sub Data	CRC16	CR
0x0A			0x47	*		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	1 Byte	4 Bytes	1 Byte

* Sub Data Details

a) In the case of SectorDoor

0x31: Region-1 (Sensor 8, 9, 11, 12)

0x32: Region-2 (Sensor 7, 8, 9, 11, 12, 13)

0x33: Region-3 (Sensor 4, 7, 8, 9, 11, 12, 13, 16)

0x34: Region-4 (Sensor 1, 4, 7, 8, 9, 11, 12, 13, 16, 19)

0x35: Region-5 (Sensor 4, 6, 7, 8, 9, 11, 12, 13, 14, 16)

0x36: Region-6 (Sensor 1, 4, 6, 7, 8, 9, 11, 12, 13, 14, 16, 19)

b) In the case of SwingDoor – bracket is region of tailing-cut

0x31: Region-1

Entry Pass : Sensor 9, 23, 24, 19, 18, 17, 16

(Normal: 24, 19, 18, Wide: 24, 19, 18, 17, 16)

Exit Pass : Sensor 6, 7, 8, 9, 23, 24, 19

(Normal: 23, 9, 8, Wide: 23, 9, 8, 7, 6)

0x32: Region-2

Entry Pass : Sensor 9, 23, 24, 19, 18, 17, 16

(Normal: 24, 19, 18, 17, 16, Wide: 24, 19, 18, 17, 16, 15)

Exit Pass : Sensor 6, 7, 8, 9, 23, 24, 19

(Normal: 23, 9, 8, 7, 6, Wide: 23, 9, 8, 7, 6, 5)

0x33: Region-3

Entry Pass : Sensor 9, 23, 24, 19, 18, 17, 16

Exit Pass : Sensor 6, 7, 8, 9, 23, 24, 19

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(2) Response

STX	Seq No.	Retry No.	DATA		Error Check Code	ETX
LF	Sequence	Retry	CMD	Error Code	CRC16	CR
0x0A			0x47	00		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes	4 Bytes	1 Byte

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4.3.13 DateTime Setting(Used only in India)

Set Date and Time of RTC(Real Time Clock) in GCU board.

(1) Command

STX	Seq No.	Retry No.	DATA			Error Check Code	ETX
LF	Sequence	Retry	CMD	Sub Data	Parameters	CRC16	CR
0x0A			0x58	*	*		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	1 Byte	0 or 12 Bytes	4 Bytes	1 Byte

* Sub Data Details

0x30: Parameter Set(Parameters length 12 Bytes)

0x31: Parameter Get(Parameters length zero)

* Paramter 1 <Date> - 6 Bytes (format: yyMMdd)

Ex 1) March 2, 2020 = 200302(0x32, 0x30, 0x30, 0x33, 0x30, 0x32)

* Paramter 2 <Time> - 6 Byte (format: HHmmss)

Ex 1) 09:44:20 am = 094420(0x30, 0x39, 0x34, 0x34, 0x32, 0x30)

Ex 2) 02:44:20 pm = 144420(0x31, 0x34, 0x34, 0x34, 0x32, 0x30)

(2) Response

STX	Seq No.	Retry No.	DATA				Error Check Code	ETX
LF	Sequence	Retry	CMD	Error Code	Sub Data	Parameters	CRC16	CR
0x0A			0x50	00	*	*		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes	1 Byte	12 Bytes	4 Bytes	1 Byte

* Sub Data

Same as Sub Data format of Command

* Parameters

Same as 12 Bytes format of Command parameter

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4.3.14 UPS(Used only when connected to UPS in India.)

Set UPS Shutdown delay time in case of AC Fail. The UPS will be truned off after the set time in AC Fail.

(1) Command

STX	Seq No.	Retry No.	DATA		Error Check Code	ETX
LF	Sequence	Retry	CMD	Sub Data	CRC16	CR
0x0A			0x59	*		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes	4 Bytes	1 Byte

* Sub Data <UPS Shutdown delay time> - 2 Bytes

Set up delay time of UPS power off when AC Failure. This command saves the set value in the GCU, so you only need to set it once. There is no need to set every AC Failure.

- Value Setting
- Range : 0x00 ~ 0xFF (0x00 : Power off immediately)
- 1 value per time : 10 secs
- Consist of 2 Byte

: if user wants to set 100 sec, then value should be set as 0x0A

(2) Response

STX	Seq No.	Retry No.	DATA		Error Check Code	ETX
LF	Sequence	Retry	CMD	Error Code	CRC16	CR
0x0A			0x59	00		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes	4 Bytes	1 Byte

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4.3.15 Standby mode setting(Used only in Kolkata, India.)

Setting the mode of GCU to switch when communication between Gate Application and GCU is disconnected. If there is no command request from the application for a period of time set by the parameters in this command, the GCU switches to the set mode.

(1) Command

STX	Seq No.	Retry No.	DATA			Error Check Code	ETX
LF	Sequence	Retry	CMD	Sub Data	Parameters	CRC16	CR
0x0A			0x55	*	*		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	1 Byte	8 or 4 Bytes	4 Bytes	1 Byte

* Sub Data 1 Details

0x30: Parameter Write

0x31: Parameter Read

* Parameter 1 <2Bytes>

0x32 0x34 (Always)

* Parameter 2 <2Bytes>

0x30 0x32 (Always)

(The items below are when Parameter Write only)

* Parameter 3 <Timeout value, 2Bytes>

0x30 0x30: default value(Not used timeout)

ex) 20 seconds: 0x31 0x34, 10 seconds: 0x30 0x3A

* Parameter 4 <Pass mode, 2Bytes>

0x33 0x3F: default value(Out of Service), range 0x33 30 ~ 0x33 0x3F (Same as passmode, set to default value when out of range)

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(2) Response

STX	Seq No.	Retry No.	DATA				Error Check Code	ETX
LF	Sequence	Retry	CMD	ErrorCode	Sub Data	Parameters	CRC16	CR
0x0A			0x55	00	*	*		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes	1 Byte	0 or 8 Bytes	4 Bytes	1 Byte

* Sub Data Details

0x30: Parameter Write

0x31: Parameter Read

(The items below are when Parameter Read only)

* Parameter 1 <Timeout value, 2Bytes>

0x30 0x30: default value(Not used timeout)

ex) 20 seconds: 0x31 0x34, 10 seconds: 0x30 0x3A

* Parameter 2 <Pass mode, 2Bytes>

0x33 0x3F: default value(Out of Service), range 0x33 30 ~ 0x33 0x3F (Same as passmode, set to default value when out of range)

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4.3.16 Open/Close Delay time setting

Setting Door Open/Close delay time.

(3) Command

STX	Seq No.	Retry No.	DATA			Error Check Code	ETX
LF	Sequence	Retry	CMD	Sub Data	Parameters	CRC16	CR
0x0A			0x55	*	*		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	1 Byte	8 or 4 Bytes	4 Bytes	1 Byte

* Sub Data 1 Details

0x30: Parameter Write

0x31: Parameter Read

* Parameter 1 <2Bytes>

0x31 0x31 (Always)

* Parameter 2 <2Bytes>

0x30 0x32 (Always)

(The items below are when Parameter Write only)

* Parameter 3 <DoorOpen delay time value, 2Bytes>

0x30 0x30: default value(Not used DoorOpen delay time)

Range : 0x30 0x30 ~ 0x30 0x3A(Per 0.1 sec / Max : 1 second)

ex) 10 seconds: 0x30 0x3A

* Parameter 4 <DoorClose delay time value, 2Bytes>

0x30 0x30: default value(Not used DoorClose delay time)

Range : 0x30 0x30 ~ 0x30 0x3A(Per 0.1 sec / Max : 1 second)

ex) 10 seconds: 0x30 0x3A

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(4) Response

STX	Seq No.	Retry No.	DATA				Error Check Code	ETX
LF	Sequence	Retry	CMD	ErrorCode	Sub Data	Parameters	CRC16	CR
0x0A			0x55	00	*	*		0x0D
1 Byte	2 Bytes	1 Byte	1 Byte	2 Bytes	1 Byte	0 or 8 Bytes	4 Bytes	1 Byte

* Sub Data Details

0x30: Parameter Write

0x31: Parameter Read

(The items below are when Parameter Read only)

* Parameter 1 <DoorOpen delay time value, 2Bytes>

0x30 0x30: default value(Not used DoorOpen delay time)

Range : 0x30 0x30 ~ 0x30 0x3A(Per 0.1 sec / Max : 1 second)

ex) 10 seconds: 0x30 0x3A

* Parameter 2 <DoorClose delay time value, 2Bytes>

0x30 0x30: default value(Not used DoorClose delay time)

Range : 0x30 0x30 ~ 0x30 0x3A(Per 0.1 sec / Max : 1 second)

ex) 10 seconds: 0x30 0x3A

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4.4 Error Code List

Code	Description	Detail
01	Sensor 01 Fail	This code appears when bootup
02	Sensor 02 Fail	This code appears when bootup
03	Sensor 03 Fail	This code appears when bootup
04	Sensor 04 Fail	This code appears when bootup
05	Sensor 05 Fail	This code appears when bootup
06	Sensor 06 Fail	This code appears when bootup
07	Sensor 07 Fail	This code appears when bootup
08	Sensor 08 Fail	This code appears when bootup
09	Sensor 09 Fail	This code appears when bootup
10	Sensor 10 Fail (Child Detect1)	This code appears when bootup
11	Sensor 11 Fail	This code appears when bootup
12	Sensor 12 Fail	This code appears when bootup
13	Sensor 13 Fail	This code appears when bootup
14	Sensor 14 Fail	This code appears when bootup
15	Sensor 15 Fail	This code appears when bootup
16	Sensor 16 Fail	This code appears when bootup
17	Sensor 17 Fail	This code appears when bootup
18	Sensor 18 Fail	This code appears when bootup
19	Sensor 19 Fail	This code appears when bootup
20	Sensor 20 Fail (Child Detect2)	This code appears when bootup
21	Sensor 21 Fail (Child Detect3)	This code appears when bootup
22	Sensor 22 Fail (Child Detect4)	This code appears when bootup
23	Sensor 23 Fail (Only used SwingDoor)	This code appears when bootup
24	Sensor 24 Fail (Only used SwingDoor)	This code appears when bootup
39	Wrongway Fraud Entry	
40	Wrongway Fraud Exit	
41	Intrusion Entry	

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42	Intrusion Exit	
43	Piggy Tailing Entry	
44	Piggy Tailing Exit	
50	Exit Timeout Entry Side	
51	Exit Timeout Exit Side	
52	Door 1 Open Fail	
53	Door 1 Close Fail	
54	Door 2 Open Fail	
55	Door 2 Close Fail	
56	CRC Error	This code appears when CMD CRC is not correct
60	Tag full error	
61	Door1 Forced Open Error (In the case of SwingDoor)	
62	Door2 Forced Open Error (In the case of SwingDoor)	
63	Door1 Over Current (In the case of SwingDoor)	
64	Door2 Over Current (In the case of SwingDoor)	
65	Door1 Over Heating (In the case of SwingDoor)	
66	Door1 Over Heating (In the case of SwingDoor)	
67	Door1 Following Error (In the case of SwingDoor)	
68	Door1 Following Error (In the case of SwingDoor)	
69	Door1 Communication Error (In the case of SwingDoor)	
70	Door2 Communication Error (In the case of SwingDoor)	
71	UPS Communication Error (For UPS use)	

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Appendix A – CRC-16

This chapter describes the checksum calculation method of CRC-16. The gcu protocol uses 16-bit CRC(Cyclic Redundancy Check) to check message transmission errors. The CRC calculation is the same as the XModem CRC calculation method. The XMODEM CRC is CRC-16 with a start value of 0x0000, the end value is not XORed(Exclusive OR), and uses a polynoimic of 0x1021.

- Calculation example

Message: 0xf2 0x00 0x03 0x43 0x30 0x30

CRC-16 calculation: 0xf61b

Send Message: 0x0a 0xf2 0x00 0x03 0x43 0x30 0x30 0x46 0x36 0x31 0x42 0x0d

STX(LF): 0x0a

Message: 0xf2 0x00 0x03 0x43 0x30 0x30

CRC-16: 0x46 0x36 0x31 0x42 <- 0xf61b

ETX(CR): 0x0d

- Reference source code

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define POLYNOMIAL      0x1021

static unsigned short crc_table[256];
unsigned short calc_crc(unsigned short crc_accum,
    unsigned char *data_blk_ptr,unsigned short data_blk_size);

void gen_crc_table ()
{
    register unsigned short i, j;
    register unsigned short crc_accum;

    for (i = 0; i < 256; i++)
    {
        crc_accum = ((unsigned short)i << 8);
        for (j = 0; j < 8; j++)
        {
```

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```

        if (crc_accum & 0x8000L)
            crc_accum = (crc_accum << 1) ^ POLYNOMIAL;
        else
            crc_accum = (crc_accum << 1);
    }
    crc_table[i] = crc_accum;
}

}

unsigned short calc_crc(unsigned short crc_accum,
    unsigned char *data_blk_ptr, unsigned short data_blk_size)
{
    register unsigned short i, j;

    for (j = 0; j < data_blk_size; j++)
    {
        i = ((unsigned short)(crc_accum >> 8) ^ *data_blk_ptr++) & 0xff;
        crc_accum = (crc_accum << 8) ^ crc_table[i];
    }

    return crc_accum;
}

/****      CRC-16 test sample      ****/
void main ()
{
    unsigned short    crc;
    unsigned char     in_frame[6] = {0xf2 0x00 0x03 0x43 0x30 0x30};

    gen_crc_table ();

    /* Calc CRC */
    crc = calc_crc (0, in_frame, 6);
    printf("\nCalc CRC result = %X", crc);
}

```